

PATENT APPLICATION: CogniClass AI

TITLE PAGE

PROJECT TITLE:

SYSTEMS AND METHODS FOR REAL-TIME AI-DRIVEN CLASSROOM INTELLIGENCE, ARTIFACT GENERATION, AND GAMIFIED ENGAGEMENT TRACKING

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1. FIELD OF THE INVENTION

The present invention relates generally to the field of educational technology, specifically to systems and methods for real-time pedagogical assistance. It concerns the integration

of multimodal Large Language Models (LLMs) with low-latency streaming protocols to facilitate automated classroom documentation, student engagement tracking, and real-time instructor feedback.

2. BACKGROUND OF THE INVENTION

In the current state of digital education, several critical gaps exist:

1. **The Transcription Gap:** Existing Speech-to-Text (STT) tools provide raw transcripts that lack semantic structure, making them difficult for students to use as study materials.
2. **The Feedback Latency:** Instructors have no objective, real-time measure of classroom "confusion" or "disengagement" in a virtual or hybrid setting.
3. **The Engagement Gap:** Traditional Learning Management Systems (LMS) are static repositories of data and lack the interactive, reward-based motivation found in modern digital ecosystems.
4. **Data Fragmentation:** Notes, quizzes, and engagement metrics are often siloed in different applications, increasing the cognitive load for both students and teachers.

The present invention seeks to solve these problems by providing a unified, AI-native intelligence layer that functions as an "Active Copilot" for the classroom.

3. SUMMARY OF THE INVENTION

The present invention provides a computer-implemented platform (CogniClass AI) comprising:

- **A Real-Time AI Orchestration Module:** Configured to process live audio streams and extract structured pedagogical metadata.
- **A Dynamic Artifact Engine:** Which converts AI-extracted metadata into structured Markdown notes and 3D interactive flashcards.
- **A Silent Help Signal (SHS) Interface:** A non-verbal student feedback mechanism that aggregates classroom sentiment into an instructor-facing heatmap.
- **A Gamified Participation Framework:** Which utilizes a persistent database to track student XP, levels, and achievements based on real-time classroom interactions.

4. BRIEF DESCRIPTION OF THE DRAWINGS (TEXTUAL)

- **FIG. 1:** System Architecture Diagram showing the flow from Client Audio -> Gemini AI -> Structured Artifacts.
- **FIG. 2:** Flowchart of the "Sliding Context Window" processing used for real-time transcription.
- **FIG. 3:** User Interface Mockup of the "Teacher Dashboard" featuring the Real-Time Confusion Heatmap.
- **FIG. 4:** Logic Diagram of the "XP Reward Engine" illustrating the relationship between student actions and database updates.
- **FIG. 5:** Cross-platform synchronization flow for PWA and Mobile devices.

5. DETAILED DESCRIPTION OF THE INVENTION

5.1 Real-Time Audio Capture and Buffer Management

In a preferred embodiment, the system utilizes the `navigator.mediaDevices.getUserMedia` API to capture high-fidelity audio. This audio is converted to a text stream via the Web Speech API. To maintain context, the system employs a "Circular Buffer" where the last 5 minutes of transcript are always available to the AI Engine for any new query, ensuring that summaries are not fragmented.

5.2 AI Semantic Categorization

The core innovation lies in the AI processing prompt. Unlike generic summaries, the present invention instructs the AI to categorize content into specific **Pedagogical Objects**:

- **OBJECT_CONCEPT:** Core terminology defined during the lecture.
- **OBJECT_PROCEDURE:** Step-by-step instructions (e.g., in the Code Lab).
- **OBJECT_QUERY:** Questions raised by students that require instructor follow-up.
- **OBJECT_SUMMARY:** Periodic "Check-In" points to summarize the last 15 minutes.

5.3 3D Interactive Flashcard Synthesis

The generation module translates **OBJECT_CONCEPT** data into a 3D-flippable card format. Each card is assigned a "Retention Score" based on how frequently the concept was

mentioned in the lecture. This data is then rendered using CSS 3D transforms, providing a tactile, immersive study experience for students at CGC University.

5.4 Aggregate Confusion Heatmap (ACH)

The ACH is a specialized instructor-side visualization. When a student clicks the "Doubt" button, a Socket.io event is fired with a timestamp and a "Confidence Score." The server aggregates these signals over a 60-second moving window. If more than 15% of the active student pool signals confusion, the instructor's dashboard highlights the specific topic currently being discussed in red, signaling a need for clarification.

5.5 Gamification and Persistent Identity

All student interactions are tied to a unique "CogniClass Identity" (verified via Email/Phone). XP is awarded not just for attendance, but for "Engagement Velocity"—the rate at which a student interacts with the AI Copilot and Code Lab.

6. CLAIMS

We claim:

- 1. A computer-implemented system for real-time classroom intelligence, comprising:**
 - a. at least one processor and a memory;
 - b. a capture module configured to receive a live audio stream from a source;
 - c. an AI orchestration module configured to perform semantic chunking and labeling on a transcription of said audio stream in real-time;
 - d. a generation engine configured to produce a plurality of interactive 3D study artifacts based on the labeled segments; and
 - e. a feedback dashboard configured to display an aggregate measure of audience comprehension.
- 2. The system of claim 1**, wherein the AI orchestration module utilizes a context window of at least 1,000,000 tokens to maintain semantic coherence across the entire duration of a classroom session.
- 3. The system of claim 1**, further comprising a gamification engine that assigns numeric values (Experience Points) to student accounts based on real-time triggers detected by the AI orchestration module.

- 4. A method for generating structured pedagogical artifacts, the method comprising:**
- a. Capturing a continuous stream of verbal lecture data;
 - b. Categorizing segments of said data into specific pedagogical objects including concepts, procedures, and queries;
 - c. Mapping each identified concept to a dual-sided 3D virtual object; and
 - d. Exporting a structured Markdown summary containing the identified procedures and summaries.
- 5. The system of claim 1,** wherein the feedback dashboard includes a "Silent Help Signal" aggregator that prevents auditory interruption while providing real-time visual alerts to an instructor regarding group confusion levels.

7. ABSTRACT

A system and method for real-time AI-driven classroom intelligence are disclosed. The invention integrates live speech-to-text processing with a generative AI engine (Gemini 1.5) to automate the creation of high-fidelity study notes and interactive 3D flashcards. Key features include a "Silent Help Signal" for non-intrusive student feedback and a gamified reward structure (XP/Badges) to incentivize participation. The system ensures data persistence through a robust MongoDB backend and provides instructors with a real-time "Confusion Heatmap" to optimize pedagogical delivery.

End of Application